C Programming Assignment – Question 1

1. Cab Fare Calculation

Question

A cab service calculates the final fare based on the distance travelled and the customer's membership. The base fare is Rs.100. If the distance travelled is more than 10 km, an additional Rs.20 is charged for every kilometer beyond 10 km. Customers with a membership receive a 10% discount on the final fare. Develop a C program to calculate the amount payable.

Sample Input

```
Distance: 15
Member: true
```

Sample Output

```
Final Fare: Rs.180
```

C Program

```c
#include <stdio.h>

int main(void) {
    float distance, fare = 100.0f;
    int member;

    printf("Enter distance travelled (km): ");
    scanf("%f", &distance);

    printf("Is the customer a member? (1 = Yes, 0 = No): ");
    scanf("%d", &member);

    if (distance > 10.0f) {
        fare += (distance - 10.0f) * 20.0f;
    }

    if (member == 1) {
        fare *= 0.90f;
    }

    printf("Final Fare: Rs.%.2f\n", fare);
    return 0;
}
```

Explanation

1. Start with a base fare of Rs.100.

2. If the distance is greater than 10 km, add Rs.20 for each kilometer beyond 10 km.

3. If the customer is a member, reduce the fare by 10%.

4. Print the final fare.

Note

The assignment's sample output says Rs.180, but applying the stated 10% membership discount to a 15 km trip gives Rs.180 before discount and Rs.162 after discount. The program follows the problem statement.